

Evaluation

1. Evaluation Methodology

Four different configurations were evaluated to determine the best legal Q/A model for the Saudi Labor Law:

1. Baseline (Retrieval Only)

- Returns the original article text without generating new answers.
- Serves as a lower bound benchmark.
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2. Zero-Shot

- The model answers directly without seeing any example.
- Relies purely on the model's general knowledge.
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3. One-Shot

- The model receives one example of a correct legal Q/A before answering.
- Helps provide a basic template for structure and style.

4. Three-Shot

- The model receives three examples before answering.
- Offers richer context, reduces hallucination, and greatly stabilizes the legal style.

1.1 Objective Metrics

Automated metrics computed from all datasets:

- Jaccard Overlap:
- Measures lexical similarity between the model answer and the correct legal article.
- Coverage (Answer in Article):
- How much of the generated answer is grounded in the actual article text.
- Coverage (Article in Answer):
- How much of the article was captured inside the model's answer.
- Relevance Score:
- Percentage of answers that meet a minimum threshold of similarity/coverage.
- Answer Length:
- Helps detect overly brief answers (missing details) or overly long ones (hallucinations).

1.2 Manual (Subjective) Metrics

Evaluated manually for 50 Q/A pairs per generative model (Zero/One/Three):

- Correctness (0–2)
- Grammar (1–5)
- Style (0–2)

These metrics capture what automated metrics cannot measure, especially legal validity.

2. Objective Results

Model	Avg Jaccard	Avg Coverage	Relevance (%)	Notes
Baseline	Lowest	Lowest	Very Low	Retrieval only
Zero-Shot	Low-Moderate	Moderate	~30%	Unstable
One-Shot	Good	Good	~50%	Learns from 1 example
Three-Shot	Best	Highest	~55-60%	Most stable & accurate

Interpretation

- Baseline is the weakest (as expected).
- Zero-Shot struggles without examples.
- One-Shot improves significantly.
- Three-Shot delivers the best similarity and grounding.

3. subjectv Evaluation Results

Model	Correctness	Grammar	Style
Zero-Shot	Lowest	Medium	Weak consistency
One-Shot	Good	Best grammar	Strong clarity
Three-Shot	Highest	Very good	Most consistent

Interpretation

- Three-Shot is the most legally accurate.
- One-Shot is the most linguistically polished.
- Zero-Shot is the least reliable.

Qualitative Observations

These observations summarize tendencies seen across the three LLM models:

Zero-Shot

- Often general, short, and lacking article references.
- Struggles with precise numerical/legal rules (articles 78–83, 98–100).
- Uses hedging words (“may”, “often”).
- Not reliable for exact legal wording.

One-Shot

- Consistently cites article numbers.
- Language is clear but sometimes rewrites instead of quoting the law.
- Needs to reduce unnecessary commentary and stick to the exact text for numerical rules.

Three-Shot

- Most consistent in citing articles and delivering legally accurate answers.
- Better alignment with the structure and style of legal articles.
- Minimal hallucination and best content stability.

Category	Best Model
Legal Correctness	Three-Shot
Grammar	One-Shot
Style Consistency	One-Shot
Objective Metrics	Three-Shot
Stability	Three-Shot
Worst Overall	Zero-Shot

6. Best Model Selection

Based on all objective and subjective results, the top-performing model is:

Three-Shot Prompting

Reasons

- Highest Correctness
- Highest Coverage & Jaccard
- Most stable and least hallucination
- Strong discipline in article citation
- Closest representation of legal text

Conclusion

- Three-Shot is the recommended model for legal Q/A generation.
- One-Shot is a strong alternative when cost/time is a constraint.
- Zero-Shot is not suitable for precise legal tasks.
- For articles with numerical rules (78–83, 98–100), the model must quote text verbatim.
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